INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Lincoln Rackhouse Maryland, MD -- station KDCA, America/New_York
Building	OSM 93210344 Lincoln Rackhouse Maryland
Facade gap	not applicable -- one building, no second facade
Weather record	43,766 real hourly records from KDCA
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-02-18 (crossing)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 4. 7 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 4 unsafe hour(s) 7 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	13.330	18.0	13.890	3.052
01	mechanical	yes	switch budget	13.340	18.0	17.220	2.577
02	mechanical	yes	switch budget	13.890	18.0	18.330	2.759
03	mechanical	yes	switch budget	13.890	18.0	18.890	2.450
04	mechanical	yes	switch budget	13.890	18.0	19.440	2.185
05	mechanical	yes	switch budget	13.890	18.0	18.890	1.900
06	mechanical	yes	switch budget	14.440	18.0	13.330	1.978
07	mechanical	no	dew point	18.330	18.0	10.560	1.470

08	mechanical	no	dry-bulb	21.660	18.0	7.780	2.178
09	mechanical	no	dry-bulb	24.440	18.0	8.330	2.888
10	mechanical	no	dry-bulb	26.670	18.0	7.780	3.094
11	mechanical	no	dry-bulb	28.330	18.0	7.220	4.078
12	mechanical	no	dry-bulb	23.330	18.0	7.220	3.877
13	mechanical	no	dry-bulb	20.560	18.0	7.780	3.989
14	FREE	yes	-	16.670	18.0	7.220	3.755
15	FREE	yes	-	15.550	18.0	6.110	3.309
16	FREE	yes	-	13.330	18.0	5.560	3.304
17	FREE	yes	-	10.550	18.0	3.890	2.800
18	FREE	yes	-	8.890	18.0	3.330	2.772
19	FREE	yes	-	7.780	18.0	2.780	2.521
20	FREE	yes	-	6.660	18.0	1.670	2.967
21	FREE	yes	-	5.000	18.0	1.110	3.007
22	FREE	yes	-	4.450	18.0	0.560	3.085
23	FREE	yes	-	3.280	18.0	0.000	3.394

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.340 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.330 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.660 C against a 18.0 C limit, so it fails by 3.660 C. It would take a limit 3.660 C higher, or a bound 3.660 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.440 C against a 18.0 C limit, so it fails by 6.440 C. It would take a limit 6.440 C higher, or a bound 6.440 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.670 C against a 18.0 C limit, so it fails by 8.670 C. It would take a limit 8.670 C higher, or a bound 8.670 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.330 C against a 18.0 C limit, so it fails by 10.330 C. It would take a limit 10.330 C higher, or a bound 10.330 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.330 C against a 18.0 C limit, so it fails by 5.330 C. It would take a limit 5.330 C higher, or a bound 5.330 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.560 C against a 18.0 C limit, so it fails by 2.560 C. It would take a limit 2.560 C higher, or a bound 2.560 C tighter, to change this hour.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.670 C, which is 1.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.755 C, and the margin is measured, not chosen: 3.755 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.550 C, which is 2.450 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.310 C, and the margin is measured, not chosen: 3.310 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.330 C, which is 4.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.304 C, and the margin is measured, not chosen: 3.304 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.550 C, which is 7.450 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.800 C, and the margin is measured, not chosen: 2.800 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.771 C, and the margin is measured, not chosen: 2.771 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 10.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.521 C, and the margin is measured, not chosen: 2.521 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.660 C, which is 11.340 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.967 C, and the margin is measured, not chosen: 2.967 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.000 C, which is 13.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.007 C, and the margin is measured, not chosen: 3.007 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.450 C, which is 13.550 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.085 C, and the margin is measured, not chosen: 3.085 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.280 C, which is 14.720 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.394 C, and the margin is measured, not chosen: 3.394 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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